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MASTER'S THESIS PROPOSAL

隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

Digital economy is measured through records that do not describe the same thing. Platform revenue, e-commerce statistics and payment data are used as interchangeable measures of digital activity, yet each captures a different side of the boundary between the value a platform processes and the revenue it recognizes. I define the difference between transaction value and platform-recognized revenue as the invisible wedge and show that substituting one record for another misstates both the size of the digital economy and where it is growing. Indonesia, the largest digital market in Southeast Asia, is the empirical setting. In 2023, Tokopedia, Shopee and Grab processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, a ratio of 13.7 to 1 and a wedge of US\$40.07 billion that stays above US\$20 billion when any one platform is excluded. Between 2020 and 2025, revenue grew faster than transaction value in ten of twelve year-to-year comparisons, so platform revenue understates the level of platform commerce and overstates its growth. National statistics place 98.5 percent of Indonesia's 2023-2024 e-commerce growth outside the marketplace channel, and digital payment value grew up to eleven times as fast as e-commerce value. Evidence from eight platform businesses outside Indonesia shows the same boundary under other business models.

1. Introduction

Southeast Asia's digital economy grew from approximately US\$100 billion in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). Digital platform scale, however, is not the same thing as platform revenue. Platforms record transactions, determine participant payouts, and book their own revenue, while national statistics and payment systems record other parts of the same economic activity. These measures are used interchangeably even though they do not measure the same thing. This paper argues that doing so produces systematically wrong conclusions about the scale and growth of platform economies.

Suppose a platform processes 100 units of transaction value but recognizes only 10 as revenue. The remaining 90 are still recorded within the platform's system but do not constitute platform revenue; they are merchant receipts, driver payouts, inventory costs, taxes, and other pass-through payments. An analyst using the platform's accounts sees 10, while one measuring the commerce it processes sees 100. I call the difference the invisible wedge.

Indonesia is the empirical setting. It is the largest digital market in Southeast Asia, and several records of the same activity can be compared there. Platform companies disclose transaction and revenue information; BPS-Statistics Indonesia, the national statistical agency, reports e-commerce activity; and Bank Indonesia, the country's central bank, reports digital payments. In 2023, the platform comparison covers Shopee and Tokopedia, which together represented about 70 percent of estimated Indonesian marketplace transaction value, together with Grab, a major intermediary in delivery and other digital services. The three platforms processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, leaving a US\$40.07 billion invisible wedge that stays above US\$20 billion when any one of them is removed.

The difference also changes over time. Across twelve year-to-year comparisons between 2020 and 2025, revenue grew faster than transaction value in ten, and the median absolute gap between the two growth rates was 42 percentage points; in some years the two measures moved in opposite directions. Platform revenue therefore understates the level of platform-mediated commerce and, in most observed years, overstates its growth.

The same problem appears beyond company accounts. BPS reports that national e-commerce value grew 17.1 percent between 2023 and 2024, but about 98.5 percent of that increase occurred outside the marketplace component, and digital payment value reported by Bank Indonesia grew up to eleven times as fast as e-commerce value. Indonesia's Minister of Finance Regulation No. 37 of 2025, which introduces seller reporting and withholding through designated marketplaces, provides a further test of how platform-held transaction records enter administrative reporting.

1.1 Research question and objectives

One question drives the proposal:

How much economic activity remains invisible when digital platforms are measured through their reported revenue?

Indonesia provides the main empirical setting. Three objectives follow: first, to measure the invisible wedge across available platform histories from 2020 to 2025; second, to explain large movements using disclosed changes in monetization, customer incentives, revenue recognition, and reporting scope; and third, to compare the platform evidence with national e-commerce and payment records to determine whether they produce the same account of economic scale and growth.

A separate comparison of eight platform businesses outside Indonesia tests whether the same transaction-revenue separation appears under other business models.

1.2 Contributions

First, the study develops the invisible wedge from a one-year estimate into a measure that can be followed over time. From 2020 to 2025, revenue grows faster than transaction value in ten of twelve annual comparisons, showing that platform revenue is not a stable measure of either the level or the growth of platform-mediated commerce.

Second, I explain why the wedge moves using accounting components disclosed by the platforms themselves. Between 2022 and 2023, Tokopedia's transaction value fell 8.9 percent while net revenue from its third-party marketplace rose 53.2 percent. Lower customer incentives account for 60.6 percent of that increase in net revenue and higher gross revenue for the remainder.

Third, I extend the comparison beyond company accounts. National e-commerce statistics show that almost all of Indonesia's 2023-2024 growth occurred outside the marketplace component, while digital payment value grew up to eleven times as fast as e-commerce value. The records therefore produce different conclusions about the scale, growth, and location of digital activity.

1.3 Positioning the contribution

The contribution is a finance measurement problem before it is a tax-policy one. Platform accounts make the intermediation boundary observable, but that boundary affects the conclusions investors, researchers, statistical agencies, and policymakers draw about scale, growth, and monetization. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter, while Berg et al. (2020) show that intermediary-held digital traces contain information conventional records can miss. This paper applies those ideas to platform economies and shows that the measure chosen to describe digital activity changes the economic conclusion.

2. Literature Review and Research Gap

Digital platforms sit between economic activity and the records used to measure it. They facilitate transactions between buyers and sellers, recognize only part of that activity as their own revenue, and generate records that can also appear in payment, statistical, and administrative systems. Existing research explains each part of this process from a different angle. Taken together, it provides the foundation for asking how much activity remains invisible when platforms are measured through reported revenue.

2.1 Platform Economics and Revenue Recognition

Platform economics explains why the value of commerce processed through a platform can be much larger than the revenue the platform records. Digital platforms often connect consumers with merchants or other service providers and earn commissions, fees, advertising, logistics, or other intermediation revenue rather than the full value of every transaction they facilitate (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016).

Accounting can widen or narrow this difference. Under IFRS 15, whether a company acts as a principal or an agent affects whether it recognizes the full customer payment or only the amount it earns from arranging the transaction (International Accounting Standards Board 2014). Customer incentives, monetization, service mix, acquisitions, and changes in business scope can also move reported revenue without an equivalent change in transaction activity. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when financial information is interpreted. The literature therefore explains why a transaction–revenue gap can exist and move over time, but not how economically large that gap becomes in practice.

2.2 Digital Records and Third-Party Information

Activity outside platform revenue is not necessarily unrecorded. Merchants and service providers can be small, self-employed, or weakly represented in conventional financial records while their transactions still leave detailed records inside a digital platform. This differs from the usual shadow-economy problem, where part of the difficulty is that economic activity leaves few reliable formal records (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Digital records can contain useful economic information that conventional records miss (Berg et al. 2020), and platform-mediated activity can create additional financial and administrative traces (Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

These records also matter once they become available to government. Public-finance research shows that compliance and enforcement change when information about income or transactions is independently reported by third parties (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules (OECD 2020) and the European Union's DAC7 regime (European Union 2021) apply this principle to digital platforms. Indonesia's Minister of Finance Regulation No. 37 of 2025 follows the same logic through seller identification, transaction-linked reporting, and marketplace withholding. The relevance here is informational: transaction records outside platform revenue still exist and can become usable elsewhere.

2.3 Measuring the Digital Economy

Official measurement frameworks already distinguish between the commerce facilitated through a platform and the service provided by the platform itself. National-accounting guidance separates the buyer-seller transaction from the platform's intermediation service rather than treating the full transaction value as platform output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025).

In practice, however, gross transaction value, platform revenue, e-commerce statistics, and payment data are all used to describe digital economic activity. They measure different things. A difference that is harmless when the measures move together becomes important when they do not, because the choice of measure can change conclusions about economic scale and growth. This thesis therefore compares the records rather than assuming that one can substitute for another.

2.4 Research Gap

Existing research explains the pieces of the problem separately. Platform economics explains why transaction value can exceed revenue; accounting explains why the gap can move; research on digital records explains why activity outside revenue can still be observable; and official measurement frameworks explain why these quantities should not be treated as equivalent.

What remains less clear is how large these differences become in practice, whether they persist over time, and whether choosing one measure instead of another changes the economic conclusion. This thesis addresses that gap by measuring the invisible wedge over time, examining disclosed reasons for major movements, comparing platform accounts with national e-commerce and payment records, and using external platform evidence to test whether the same measurement problem appears beyond the main Indonesian setting.

3. Theoretical Framework and Hypotheses

Digital platform activity can be observed through several records, but those records do not measure the same economic object. The framework begins with the difference between the value a platform processes and the revenue it recognizes, then asks whether that relationship remains stable over time and whether other records produce the same account of digital activity.

For the platform analysis, transaction value and revenue are paired only when they refer to the same period and business activity. Broader-scope, reconstructed, national-statistical, and payment evidence is kept in separate analytical roles rather than combined into one sample.

3.1 Measurement Framework

Let V_{it} denote transaction value processed by platform i in year t , and R_{it} the revenue it recognizes over the same period and business scope. The absolute invisible wedge is:

$$W_{it} = V_{it} - R_{it}$$

W measures transaction value outside the platform's recognized revenue. It does not by itself imply participant profit, taxable income, value added, non-compliance, or tax due. Because platforms differ in size, I also express the wedge relative to revenue through the Ecosystem Ratio, which is constructed for this study:

$$E_{it} = (V_{it} - R_{it}) / R_{it}$$

An E of 9 means that for every one unit of platform revenue, nine additional units of transaction value lie outside the revenue line. A single ratio does not show whether transaction value and revenue move together through time, so I define annual growth divergence as:

$$D_{it} = g(V_{it}) - g(R_{it})$$

A positive D means transaction value grew faster than revenue; a negative D means revenue grew faster. Together, W , E , and D measure the size of the transaction-revenue boundary and how that boundary changes over time.

3.2 Hypotheses

The four hypotheses examine different layers of the same measurement problem. H1 tests the transaction-revenue boundary within platforms. H2 and H3 examine where activity and records outside platform revenue appear in national e-commerce evidence. H4 is the central claim of the thesis, and H1 to H3 establish the mechanisms behind it.

H1. Platform revenue is not a stable proxy for the transaction value a platform processes.

If it were, transaction value and revenue would move proportionally within the same platform over time.

H2. Aggregate e-commerce growth is driven mainly by growth in the number of participating businesses rather than by higher transaction value per business.

This separates wider business participation from changes in the average value observed per business.

H3. Marketplace participation is associated with stronger financial recordkeeping.

This asks whether marketplace-mediated activity leaves a different recordkeeping pattern from other e-commerce activity.

H4. Platform revenue, national e-commerce statistics and payment records are not interchangeable: substituting one for another changes the measured scale and growth of digital economic activity.

This asks whether the records produce the same economic account when they observe different parts of digital activity.

Table 1. Hypotheses and principal empirical tests.

Hypothesis	Principal empirical test
H1 - Transaction-revenue divergence	Platform transaction value and revenue through time, the Ecosystem Ratio, and annual growth divergence
H2 - Sources of e-commerce growth	BPS national e-commerce value and business-count decomposition
H3 - Marketplace participation and recordkeeping	BPS published marketplace and financial-recordkeeping evidence
H4 - Measurement choice	Comparison of platform, national e-commerce, and payment records

The comparison of platform businesses outside Indonesia is used as corroboration rather than as a fifth hypothesis. Indonesia's marketplace reporting regime is examined as an institutional application in Section 5.3.

4. Data and Methodology

The study uses several sources because no single record captures every part of digital platform activity. Platform reports provide transaction value and revenue; BPS-Statistics Indonesia provides national e-commerce and business evidence; Bank Indonesia provides payment statistics; and platform companies outside Indonesia provide a separate comparison. These sources answer different questions and are not combined into one dataset.

4.1 Selecting the Platform Evidence

The basic observation is one platform in one year, with transaction value and revenue referring to the same period and, as closely as possible, the same business activity. An observation is retained only when its period, geographic and business coverage, units, definitions, and source can be identified. Measures referring to different periods are excluded, while major changes in accounting definitions or business structure are recorded.

Table 2. Indonesian platform evidence.

Type of evidence	Platforms	Annual observations	Role
Direct Indonesia-aligned data	Tokopedia e-commerce	2	Main direct anchor
Company-reported data with broader scope	Blibli third-party business; Bukalapak Group	9	Evidence through time with stated scope limits
Indonesia estimates requiring construction	Grab; Shopee Indonesia	6	Supporting country evidence

An annual observation is one matched transaction-and-revenue pair for one platform in one year. The 17 retained observations produce 12 year-to-year comparisons because the same platforms are not available in every year. Tokopedia is closely aligned with Indonesia but is not a literal country line. Blibli includes online travel, Bukalapak includes some overseas activity, and Grab and Shopee require additional construction. Tokopedia 2021 and Bukalapak 2024 are excluded because their transaction and revenue figures cover different periods.

4.2 Comparing Platform Activity Through Time

Changes through time are examined mainly by comparing each platform with itself. This keeps the company, business activity, and reporting convention as similar as the disclosures allow. When transaction value and revenue move sharply apart, disclosed changes in monetization, customer incentives, revenue recognition, and business scope are used to reconcile the movement where the filings permit.

Companies also use different names for transaction activity, including gross transaction value (GTV), gross merchandise value (GMV), and total payment volume (TPV). The original definitions are retained rather than assumed to be identical.

The cross-platform comparison uses the 2023 fiscal year, the last year in which the three principal platforms can be compared on the same basis. It measures the scale of the transaction-revenue

difference for those platforms, and market-share estimates describe their coverage of marketplace transaction value.

4.3 Direct and Constructed Indonesia Measures

Not every company reports the same Indonesia-specific information.

Table 3. Construction of the principal platform measures.

Platform	Directly available	Constructed or external input	Main limitation
Tokopedia	E-commerce transaction value and segment revenue	None for matched pair	Indonesia-aligned segment, not literal country line
Grab	Indonesia revenue; Group revenue and GTV	Indonesia GTV derived from Group figures	Assumes Group revenue-to-GTV relationship applies to Indonesia
Shopee	Group service monetization rate	External Indonesia transaction estimate; Indonesia revenue derived	Country measures are not independent disclosures
Blibli / Bukalapak	Reported transaction and revenue pairs	None for retained pairs	Scope extends beyond Indonesia alone

Grab reports Indonesian revenue directly but not Indonesian transaction value. Indonesian GTV is therefore estimated as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

Sea Limited does not report Shopee transaction value separately for Indonesia. Indonesian transaction value is taken from Momentum Works, and Indonesian revenue is estimated using Sea's Group service monetization rate:

$$\text{Indonesia revenue} = \text{Indonesia transaction value} \times \text{Group monetization rate}$$

Both constructions are labeled wherever they are used, and Section 5.1 reports how far the results move under alternative values of their inputs.

4.4 Evidence Beyond Platform Accounts

Table 4. Evidence modules and their role.

Evidence source	What is observed	Role
Indonesian platform histories	Transaction value and revenue through time	Measure the wedge and its movement
BPS national e-commerce statistics	National transaction value and estimated number of businesses	Examine sources of aggregate growth
BPS marketplace study	Business-level marketplace and recordkeeping evidence	Examine the association with financial recordkeeping
Bank Indonesia	Digital-payment measures	Compare payment growth with e-commerce growth
Platform companies outside Indonesia	Matched transaction and revenue measures	Test whether the same boundary appears elsewhere

For H2, national e-commerce value is expressed as:

$$V = N \times A$$

where N is the estimated number of e-commerce businesses and A is implied nominal transaction value per business. Changes in total value are divided arithmetically between the business-count and value-per-business components. A conflicting published 2023 business count is retained as a sensitivity check.

For H3, BPS's published business-level evidence is used to examine the association between marketplace participation and financial recordkeeping, with province-level correlations as supporting diagnostics. For H4, Bank Indonesia payment measures are compared with e-commerce growth but are not added to platform or BPS transaction values.

4.5 External Evidence and Robustness

The external comparison contains 48 matched annual transaction-and-revenue observations from eight platform businesses covering 2017-2025. It tests whether the same transaction-revenue boundary appears under other business models and is kept separate from the Indonesian sample.

The robustness checks remove one platform at a time, vary every constructed input, replace ordinary growth rates with log changes, drop the most extreme year-to-year comparison, and restrict the evidence to pairs the companies report directly. Section 5 reports the resulting ranges.

5. Preliminary Evidence and Feasibility

The evidence assembled so far establishes that the proposed comparisons can be carried out and that the choice of record produces economically large differences. The preliminary results first show the scale and movement of the transaction-revenue boundary, then examine whether the wider Indonesian evidence tells the same story.

5.1 Platform Scale and Movement

The 2023 fiscal year provides a common comparison across Grab, Tokopedia, and Shopee.

Table 5. Indonesia-focused 2023 platform comparison, US\$ billions.

Case	Transaction value V	Revenue R	Wedge W	Ecosystem Ratio E	Evidence
Grab Indonesia	5.381	0.605	4.776	7.895x	Transaction value reconstructed
Tokopedia	16.331	0.405	15.926	39.296x	Direct matched pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Transaction value and revenue estimated
Three platforms	43.233	3.162	40.070	12.671x	Sum of three cases

Across the three platforms, US\$43.23 billion of transaction value corresponds to US\$3.16 billion of recognized revenue, leaving a US\$40.07 billion wedge; transaction value is 13.7 times recognized revenue. Tokopedia contributes 39.7 percent of the wedge, Shopee 48.3 percent and Grab 11.9 percent. Shopee's ratio of exactly 9.000 follows from applying Sea's Group monetization rate of 10.0 percent, so its revenue is an assumption. Varying that rate between 9 and 11 percent moves the combined wedge only between US\$39.86 billion and US\$40.29 billion, because Shopee's revenue is small relative to its transaction value. Excluding any one platform leaves a wedge between US\$20.70 billion and US\$35.29 billion, varying every constructed input one at a time keeps it between US\$37.65 billion and US\$42.49 billion, and Tokopedia alone, which requires no construction, carries a wedge of US\$15.93 billion.

The relationship between the two measures is not stable. Across twelve year-to-year comparisons, revenue grows faster than transaction value in ten, and the median absolute gap between the two growth rates is 42.02 percentage points. Restricting the evidence to pairs reported directly by Tokopedia, Blibli and Bukalapak gives nine comparisons, in which revenue grows faster in six and the median absolute gap is 42.94 points. On that restricted evidence the gap is 36.64 points in log changes and 29.34 points after removing the single largest comparison, stays between 29.34 and 52.52 points when any one comparison is left out, and every version that excludes one platform keeps at least one year in which the two measures move in opposite directions.

Tokopedia shows why the two measures separate. Between 2022 and 2023, its transaction value fell 8.9 percent while net revenue from its third-party marketplace rose 53.2 percent. Lower customer incentives account for 60.6 percent of that increase and higher gross revenue for the remaining 39.4 percent, so the platform's revenue improved while the commerce it carried contracted.

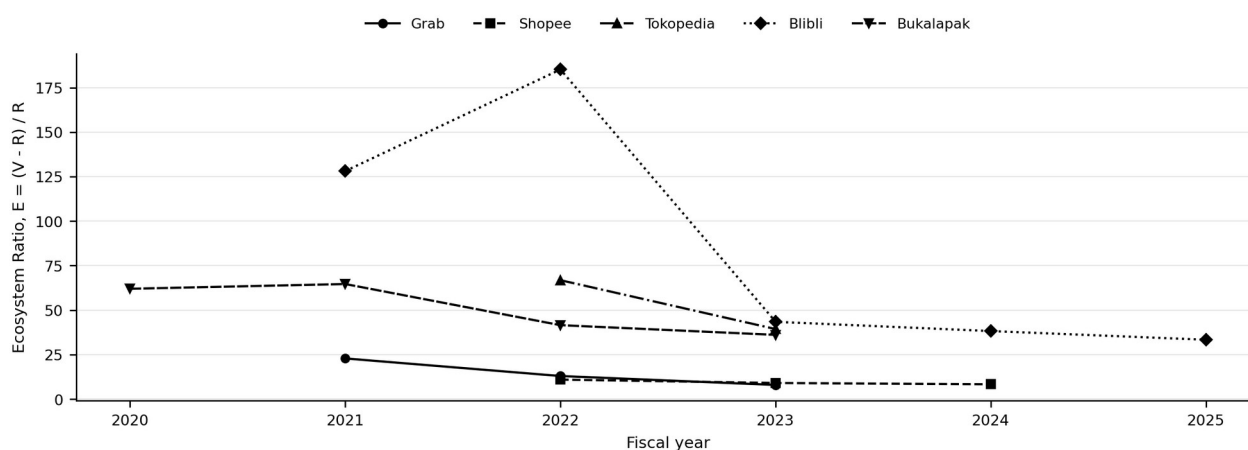


Figure 1. Ecosystem Ratio through time for each platform, 2020-2025. Each line follows one platform's matched transaction value and revenue.

5.2 Evidence Beyond Platform Revenue

National e-commerce and business evidence provides a different view of the same digital economy. Table 6 summarizes the preliminary evidence corresponding to H2-H4.

Table 6. Preliminary evidence beyond platform accounts.

Question	Preliminary evidence	Interpretation
Where does e-commerce growth come from?	E-commerce value +17.08%; estimated business count +15.31%; implied nominal value per business +1.54%. The business count accounts for 90.29% of the increase (70.95% under the conflicting count)	Growth comes mainly from more participating businesses (H2)
Where does that growth occur?	98.46% of the nominal increase occurs outside the marketplace component; marketplace share falls from 18.2% to 15.8%	Marketplace-based measurement misses almost all recent growth
Are marketplace use and financial records related?	Complete financial statements: 28.63% of marketplace users versus 12.25% of non-marketplace users	Marketplace participation is associated with stronger financial recordkeeping (H3)
Do other digital records show the same growth?	BPS e-commerce +17.08%; electronic-money shopping +30.47%; mobile-banking payments and purchases +82.84%; QRIS +186.98%	Payment measures grow 1.8 to 11 times as fast as e-commerce value (H4)

The records do not agree. Between 2023 and 2024, marketplace value grew 1.45 percent, national e-commerce value grew 17.08 percent and payment measures grew between 30 and 187 percent, so the same year reads as stagnation or as rapid expansion depending on which record is used.

5.3 Feasibility and Institutional Application

The preliminary evidence is sufficient to motivate the full analysis. H1 is supported by the movement between transaction value and revenue within platforms, H2 and H3 by national and business-level BPS evidence, and H4 by the contrast between platform, e-commerce, and payment records.

Outside Indonesia, 48 matched company-years for eight platform businesses, including eBay, Etsy, Shopify, Mercado Libre and Sea, show the same boundary under other business models, with take rates among the marketplace businesses ranging from about 3 percent at Shopify to about 25 percent at Mercado Libre. Eleven of 40 year-to-year comparisons fail a clean-scope test because of acquisitions or perimeter changes, so the comparability problem the Indonesian design manages is a general feature of platform disclosure.

PMK 37/2025 applies the same distinction in policy. It requires designated marketplace operators to identify sellers, report transaction-linked turnover and withhold tax, with implementation scheduled for 1 November 2026 after a postponement. Its reach is set by the channel it targets: the marketplace component carried 15.8 percent of 2024 e-commerce value and 1.5 percent of 2023-2024 growth.

At this stage, the proposed measure can be constructed, followed through time, reconciled to disclosed accounting changes, and compared with independent records of Indonesian digital activity.

6. Why the Problem Matters

These records are designed to measure different things, and the error arises when one is used in place of another. Platform revenue measures what the platform recognizes as its own income, transaction value measures the commerce it processes, national e-commerce statistics describe the wider market, and payment systems record the movement of funds. An investor valuing platforms on revenue growth, a statistical agency sizing e-commerce from marketplace data and a policymaker reading payment growth as commerce growth are each working from a record that departs sharply from the activity they intend to measure, and each in a different direction.

Administrative use follows the same logic. Platform-held transaction records contain seller and transaction information that never appears in corporate revenue, and they become useful to government only when identity, reporting, transmission, and matching are in place. Indonesia's new reporting regime is built around the marketplace channel, so it inherits the coverage of that channel.

7. Limitations and Robustness

Table 7. Claims, evidence and robustness tests.

Claim	Evidence and robustness test
Platform revenue understates platform commerce by more than an order of magnitude.	2023 transaction value is 13.7 times revenue (wedge US\$40.07bn). Holds across Shopee monetization of 9-11% (US\$39.86-40.29bn), with any one platform excluded (at least US\$20.70bn), under one-at-a-time input variation (US\$37.65-42.49bn), and on Tokopedia alone (US\$15.93bn).
Platform revenue growth misstates commerce growth.	Revenue outgrows transaction value in 10 of 12 year-to-year comparisons; median absolute gap 42.02 pp. Holds on directly reported pairs only (6 of 9; 42.94 pp), in log changes (36.64), without the largest comparison (29.34 pp), and with any one platform excluded.
Marketplace statistics miss where e-commerce grows.	98.46% of 2023-2024 growth in e-commerce value falls outside the marketplace component. Holds on both the published amount and the rounded-share reconstruction (98.49%).
Payment data overstate commerce growth.	Electronic-money, mobile-banking and QRIS payment value grew 1.8 to 11 times as fast as e-commerce value, 2023-2024.

The evidence is descriptive and the sample is small. The business-count decomposition is arithmetic, the recordkeeping result is an association, the Tokopedia reconciliation is an accounting identity rather than a causal estimate, and the ranges above are sensitivity ranges rather than statistical confidence intervals. Payment values are compared with e-commerce growth but are not treated as estimates of sales, and the compliance and revenue effects of PMK 37/2025 can be observed only after implementation.

Two sample decisions are put to the committee: whether Blibli and Bukalapak, whose reported figures extend beyond Indonesian goods marketplaces, enter the main longitudinal sample, and whether Grab and Shopee remain supporting evidence. The central results hold under either choice, because the growth result holds on directly reported pairs and the level result holds on Tokopedia alone.

8. Next Steps

The main data and empirical framework are already assembled. The remaining work is to settle the sample decisions, complete the robustness checks, verify source consistency, and develop the preliminary results into the full thesis analysis.

Table 8. Planned work toward the final thesis.

Period	Planned work
Proposal stage	Incorporate feedback from the proposal examination and finalize the empirical specification.
October-November 2026	Complete remaining source checks and robustness analysis; finalize the main tables and figures.
Before the final defense	Complete the full manuscript, integrate supporting evidence where it strengthens the argument, and conduct the final citation and consistency review.

The proposed study begins from a simple distinction between the value a platform processes and the revenue it recognizes. The preliminary evidence shows that this boundary is economically large, changes through time, and gives a different account of Indonesia's digital economy from national statistics and payment records. The final thesis measures how far those accounts diverge and which conclusions change when one record is used in place of another.

Appendix A - Data Sources by Evidence Type

Table A1. Source lineage by evidence type.

Evidence type	Platform or source	Transaction-value source	Revenue or comparison source
Direct Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Company-reported, broader scope	Blibli third-party business	Global Digital Niaga prospectus and results releases	Same issuer filings
Company-reported, broader scope	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Indonesia reconstruction	Grab Indonesia	Derived from Indonesia revenue and Group monetization rate	Grab Form 20-F
Indonesia reconstruction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023 and 2024; marketplace study	Business-count and recordkeeping evidence
Payment statistics	Bank Indonesia	Payment System Statistics and QRIS reports	Growth-rate comparison only
External corroboration	Eight platform businesses	Issuer transaction measures	Issuer revenue measures
Regulatory	PMK 37/2025; DJP	Ministry of Finance / Directorate General of Taxes	Institutional architecture only

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

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